

SHERBROOKE AP, QC, Canada

WMO: 716100

Lat: 45.4403N

Lon: 71.6918W

Elev: 792

StdP: 14.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.8	-11.1	-25.9	1.3	-15.7	-20.2	1.9	-10.6	26.0	24.5	22.9	18.3	4.9	270	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	84.2	70.4	81.5	68.6	78.9	67.2	72.9	80.4	71.0	78.3	69.3	76.2	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	115.0	77.3	68.5	107.8	75.3	66.7	101.3	73.2	37.1	80.3	35.4	78.5	33.9	76.3	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.1	15.8	DB	-25.6	88.8	7.0	1.7	-30.7	90.0	-34.8	91.0	-38.7	92.0	-43.8	93.2	(4)
(5)			WB	-25.9	76.2	6.9	2.0	-30.9	77.6	-35.0	78.8	-38.8	79.9	-43.9	81.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.7	13.5	16.2	26.2	39.8	53.0	61.6	66.2	64.3	57.0	45.3	33.5	21.5		
	DBStd	20.65	14.07	12.51	11.61	8.70	7.78	6.61	5.21	5.51	7.36	8.33	9.96	11.65		
	HDD50	4799	1130	945	740	325	56	2	0	0	19	197	500	885		
	HDD65	8729	1595	1365	1203	756	378	141	47	81	257	612	945	1349		
	CDD50	1753	0	0	2	19	150	351	504	443	229	50	5	0		
	CDD65	210	0	0	0	0	7	40	86	59	17	1	0	0		
	CDH74	2046	0	0	0	15	147	437	747	518	174	8	0	0		
	CDH80	389	0	0	0	2	26	95	152	83	31	0	0	0		
	WSAvg	6.4	7.2	7.3	7.6	7.5	6.6	5.6	5.2	4.9	5.1	6.3	6.8	6.7		
	PrecAvg	45.7	3.1	2.5	3.2	3.1	3.8	4.1	4.7	5.2	4.0	3.6	4.3	4.1		
(15) Precipitation	PrecMax	61.2	6.3	4.6	5.3	7.7	7.4	7.1	8.9	8.2	7.0	6.9	8.5	7.6		
	PrecMin	36.6	0.9	0.8	0.7	0.9	1.8	1.4	1.9	2.3	1.4	1.1	2.4	1.9		
	PrecStd	6.7	1.3	1.0	1.2	1.3	1.6	1.5	2.0	1.6	1.5	1.5	1.2	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	50.2	61.4	75.3	83.8	86.9	87.7	85.8	84.1	74.9	65.3	50.8		
		MCWB	49.2	43.2	50.6	58.0	66.0	72.1	74.1	72.2	71.2	61.9	57.6	47.1		
	2%	DB	42.9	42.9	51.9	67.8	78.2	82.6	84.0	82.3	78.9	69.0	58.4	45.5		
		MCWB	40.7	38.3	43.6	54.2	61.3	69.3	71.4	69.6	67.9	59.1	51.9	43.1		
	5%	DB	37.4	38.3	46.4	61.0	73.5	79.3	81.0	79.6	74.9	64.3	53.0	40.9		
		MCWB	35.4	34.8	39.8	49.7	60.0	67.0	69.2	68.2	65.3	56.8	47.6	39.3		
	10%	DB	33.3	34.7	42.1	55.2	68.8	75.5	78.3	76.7	70.8	59.5	48.1	36.5		
		MCWB	31.4	32.0	37.1	45.8	57.1	64.9	67.7	66.8	63.3	53.5	44.2	34.5		
	0.4%	WB	49.0	45.1	53.9	61.9	68.4	74.4	76.0	73.9	72.6	65.2	58.8	47.7		
		MCDB	51.8	48.7	61.7	72.3	78.7	83.6	85.1	82.3	81.3	72.0	62.8	50.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.2	39.3	45.3	55.9	65.0	71.4	73.4	71.6	69.5	61.1	53.3	43.4		
		MCDB	42.6	42.0	49.6	64.8	73.6	79.6	80.0	78.8	76.3	66.5	57.6	45.0		
	5%	WB	35.4	35.1	40.8	51.1	61.9	68.8	71.4	69.8	67.0	57.6	48.7	39.2		
		MCDB	37.3	37.6	45.6	58.1	70.2	76.4	78.3	76.8	72.8	63.3	52.0	41.0		
	10%	WB	31.6	32.3	37.6	47.1	58.7	66.5	69.4	68.1	64.3	54.2	44.4	34.7		
		MCDB	33.2	34.6	41.6	54.5	66.0	73.2	76.1	74.5	69.5	58.8	48.0	36.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	20.0	19.0	19.4	22.3	21.4	20.9	21.4	21.9	18.7	15.3	15.2		
		MCDBR	20.0	20.6	25.0	31.4	30.2	27.2	24.2	25.2	26.7	27.1	23.2	16.8		
		MCWBR	18.8	16.9	17.5	19.1	15.9	14.9	12.9	13.8	16.1	17.5	17.8	15.8		
		MCDBR	19.7	19.1	21.4	27.1	24.5	22.9	20.9	21.5	23.0	23.9	20.5	16.0		
	5% WB	MCWBR	19.0	16.5	16.3	18.0	14.4	13.8	12.1	12.7	14.7	17.0	17.3	15.6		
(40) Clear-Sky Solar Irradiance	taub		0.264	0.282	0.311	0.345	0.369	0.410	0.416	0.396	0.364	0.326	0.302	0.274		
	taud		2.371	2.287	2.337	2.390	2.387	2.267	2.257	2.317	2.405	2.510	2.496	2.392		
	Ebn at Noon		272	286	292	290	285	272	269	271	271	268	256	254		
	Edh at Noon		23	31	34	36	37	42	42	38	32	25	21	21		
(44) All-Sky Solar Radiation	RadAvg		442	722	1106	1380	1553	1724	1720	1528	1238	715	434	308		
	RadStd		62	90	83	153	175	131	148	93	104	69	57	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+2.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page